Chemistry 1 Kimia 1

Session 2026/2027 Semester 1

CHEMISTRY UNIT

KOLEJ MATRIKULASI MELAKA

SMARTCHEM

Set 1

CHEMISTRY 1

2 hours

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU

DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO

INSTRUCTIONS TO CANDIDATE:

This question paper consists of 6 questions.
Answer all questions.

All steps must be shown clearly. Use a new page for each question.

Maximum marks awarded are shown in brackets at the end of each question or section. The use of electronic calculator is permitted.

Kertas soalan ini mengandungi **6** halaman bercetak.

*This booklet consists of **6** printed pages.*

TABLE OF RELATIVE ATOMIC MASSES

Element	Symbol	Proton number	Relative atomic mass
Aluminium	Al	13	27.0
Silver	Ag	47	107.9
Argon	Ar	18	40.0
Arsenic	As	33	74.9
Gold	Au	79	197.0
Barium	Ba	56	137.3
Beryllium	Be	4	9.0
Bismuth	Bi	83	209.0
Boron	B	5	10.8
Bromine	Br	35	79.9
Iron	Fe	26	55.9
Fluorine	F	9	19.0
Phosphorus	P	15	31.0
Helium	He	2	4.0
Mercury	Hg	80	200.6
Hydrogen	H	1	1.0
Iodine	I	53	126.9
Cadmiun	Cd	48	112.4
Potassium	K	19	39.1
Calcium	Ca	20	40.1
Carbon	C	6	12.0
Chlorine	Cl	17	35.5
Cobalt	Co	27	58.9
Krypton	Kr	36	83.8
Chromium	Cr	24	52.0
Copper	Cu	29	63.6
Lithium	Li	3	6.9
Magnesium	Mg	12	24.3
Manganase	Mn	25	54.9
Sodium	Na	11	23.0
Neon	Ne	10	20.2
Nickel	Ni	28	58.7
Nitrogen	N	7	14.0
Oxygen	O	8	16.0
Platinum	Pt	78	195.1
Lead	Pb	82	207.2
Protactinium	Pa	91	231.0
Radium	Ra	88	226.0
Radon	Rn	86	222.0
Rubidium	Rb	37	85.5
Selenium	Se	34	79.0
Cerium	Ce	58	140.1
Cesium	Cs	55	132.9
Silicon	Si	14	28.1
Scandium	Sc	21	45.0
Tin	Sn	50	118.7
Antimony	Sb	51	121.8
Strontium	Sr	38	87.6
Sulphur	S	16	32.1
Uranium	U	92	238.0
Tungsten	W	74	183.9
Xenon	Xe	54	131,3
Zinc	Zn	30	65.4

LIST OF SELECTED CONSTANT VALUES

Ionisation constant for water at 25°C	K_w	= 1.00 x 10^{-14} mol^2 dm^{-6}
Molar volume of gases	V_m	= 22.4 dm^3 mol^{-1} at STP = 24 dm^3 mol^{-1} at room temperature
Speed of light in a vacuum	c	= 3.0 x 10^8 m s^{-1}
Specific heat of water		= 4.18 kJ kg^{-1} K^{-1} = 4.18 J $g^{-}1$ K^{-1} = 4.18 J $g^{-1}°C^{-1}$
Avogadro's number	N_A	= 6.02 x 10^{23} mol^{-1}
Faraday constant	F	= 9.65 x 10^4 C mol^{-1}
Planck constant	h	= 6.6256 x 10^{-34} J s
Rydberg constant	R_H	= 1.097 x 10^7 m^{-1} = 2.18 x 10^{-18} J
Molar of gases constant	R	= 8.314 J K^{-1} mol^{-1} = 0.08206 L atm mol^{-1} K^{-1}
Density of water at 25°C		= 1 g cm^{-3}
Freezing point of water		= 0.00 °C
Vapour pressure of water at 25°C	P_{water}	= 23.8 torr

UNIT AND CONVERSION FACTOR

VOLUME	1 liter = 1 dm^3 1 mL = 1 cm^3
ENERGY	1 J = 1 kg m^2s^{-2} = 1 N m = 1 x 10^7 erg 1 calorie = 4.184 J 1 eV = 1.602 x 10^{-19} J
PRESSURE	1 atm = 760 mmHg = 760 torr =101.325 kPa = 101 325 N m^{-2}
OTHERS	1 faraday (F) = 96 500 coulomb 1 newton (N) = 1 kg m s^{-2}

1. a) **FIGURE 1** shows the mass spectrum of magnesium, Mg which has 12 protons.

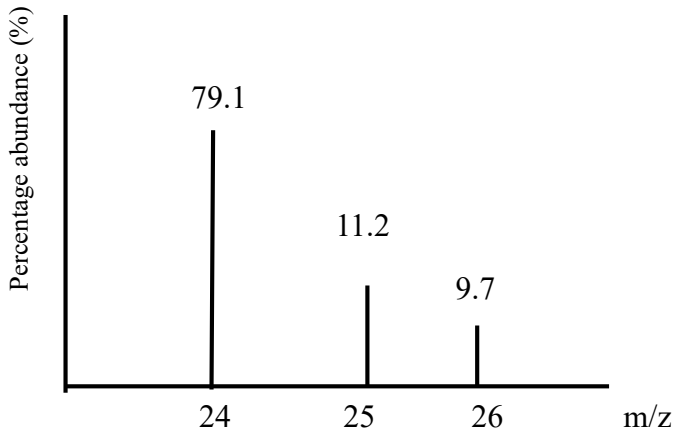


FIGURE 1

(i) Determine the relative atomic mass of magnesium.

(ii) Write the isotopic notation of the most abundant magnesium ion formed when a magnesium atom undergoes ionization by losing two electrons.

[4 marks]

b) An organic compound Y consists of carbon, hydrogen and oxygen atoms. The combustion of 1.18 g sample of Y yields 1.76 g CO_2 and 0.54 g H_2O.

Determine the empirical formula of compound Y.

[6 marks]

c) A solution is prepared by dissolving 9.660 g of thiophene, C_4H_4S in 260.0 mL of toluene, C_7H_8. The density of toluene is 0.867 g mL^{-1}. Calculate the molality of thiophene in the solution.

[4 marks]

d) At room temperature, sulphur dioxide gas, SO_2 and water, H_2O are produced by the reaction of hydrogen sulphide gas, H_2S with oxygen gas, O_2 according to the equation below:

$$2H_2S\ (g)\ +\ 3O_2\ (g) \longrightarrow 2SO_2\ (g) + 2H_2O\ (\ell)$$

(i) Determine the mass of SO_2 produced when 6.00 dm^3 of H_2S reacts with 12.0 dm^3 of oxygen gas.

(ii) Calculate the mass of the excess reactant remaining after the reaction.

[7 marks]

2. a) When an electron makes a transition from higher energy level to a lower energy level, a photon with a wavelength of 2166 nm is released.

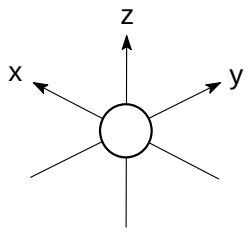
(i) Calculate the energy of the photon.

(ii) Determine the line in Brackett series that corresponds to this transition.

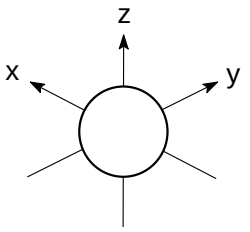
(iii) Sketch the energy level diagram to illustrate the electron transition that produces the line in 2a)(ii).

[7 marks]

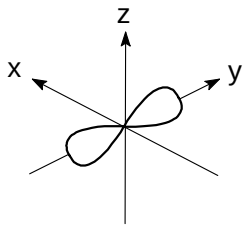
b) The orbitals of the first two energy levels of an atom are shown below.



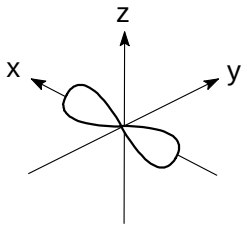
A



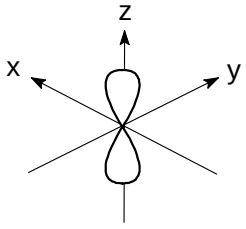
B



C



D



E

In an atom J, the orbitals A, B and C are full of electrons while orbitals D and E are half full.

(i) Draw the orbital diagram of atom J.

(ii) State the most stable ion of J.

(iii) Write a set of quantum number for an electron in orbital D.

[3 marks]

3. a) Aluminium is one of the widely used metals found in cookware, window frames, and versatile foil paper.

(i) Draw the most stable Lewis structure of aluminium chloride, $AlCl_3$.

(ii) The central atom aluminium in an aluminium chloride, $AlCl_3$ molecule does not obey the octet rule. State the name of the process where two molecules of $AlCl_3$ react to satisfy the octet rule. Draw the Lewis structure of the species formed and state the type of chemical bond formed between two $AlCl_3$ molecules.

(iii) Among the three metals in Period 3, the boiling point of aluminium, Al, is higher than that of magnesium, Mg, and sodium, Na. State two factors that contribute to the higher boiling point of Al.

[6 marks]

b) Silicon, Si, and phosphorus, P, are Period 3 elements with significantly different boiling points. The boiling point of Si is 3265°C, whereas the boiling point of P is 280.5°C. Explain this phenomenon.

[4 marks]

c) A sulphur and chlorine atom can form two neutral molecules, sulphur dichloride, SCl_2, and sulphur tetrachloride, SCl_4, which exhibit different stabilities at room temperature.

(i) Draw the molecular shape of SCl_2 and SCl_4. Then, state the approximate Cl-S-Cl bond angle in each molecule.

(ii) Both SCl_2 and SCl_4 are polar molecules. Explain.

[7 marks]

4. a) A piece of sodium metal reacts completely with excess hydrogen chloride gas as follows:

$$2Na(s) + 2HCl(g) \rightarrow 2NaCl(aq) + H_2(g)$$

Hydrogen gas produced in the reaction is collected over water at 25°C. If 320 mL of gas is collected at a pressure of 0.980 atm, calculate the mass of hydrogen produced in the reaction.

[5 marks]

b) The normal melting point and boiling point of a pure substance ***X*** are 200 K and 263 K respectively. Its triple point is at 198 K and 1.7 x 10^{-3} atm, whereas critical point is at 430 K and 78 atm. Solid ***X*** has smaller volume than liquid ***X***.

(i) Sketch the phase diagram of ***X***.

(ii) Based on the phase diagram in b)(i), state the pressure at which solid X undergoes sublimation.

[4 marks]

5. a) A sample of ammonium hydrogen carbonate is heated in a closed container at 100°C until equilibrium is established.

$$NH_4HCO_3(s) \rightleftharpoons NH_3(g) + H_2O(g) + CO_2(g)$$

If the equilibrium constant, K_p for the reaction is 0.0640, calculate the total pressure of the gases (atm) present in the container at equilibrium.

[5 marks]

b) A 5.00-L flask contains 1.00 mol of N_2, 2.00 mol of H_2 and 0.500 mol of NH_3 at 500°C. Determine the direction of reaction to reach equilibrium.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \qquad K_c = 0.250$$

[4 marks]

6. a) Methylamine, CH_3NH_2, is a weak base. A solution contains 0.0020 mol dm^{-3} CH_3NH_2 and dissociates by 0.20%. Calculate the pH of the solution.

[5 *marks*]

(b) A buffer solution is prepared by mixing 0.20 M nitrous acid, HNO_2, with 0.10 M sodium nitrite, $NaNO_2$, solution. Write the chemical equations when a small amount of base is added to the solution. Explain the effect on the pH of the solution

[*3 marks*]

(c) A sample of acetic acid, CH_3COOH, has a pH of 2.78. It needs 25 mL of 0.2 M NaOH to reach the equivalence point. Sketch the titration curve. Label the buffer region and the equivalence point on the graph.

[2 marks]

(d) Solid AgCl is placed in a beaker of water. After some time, the concentrations of Ag^+ ions and Cl^- ions are measured and found to be 1.0×10^{-6} M each. Will a precipitate form in this situation?
[K_{sp} of AgCl = 1.0×10^{-10}]

[4 *marks*]

End of questions